Chapter 12 — Human Skeletal and Muscular Systems

Q.1 — MULTIPLE CHOICE QUESTIONS (8 × 1 = 8)

1. Membrane covering bone surface is: (a) Perichondrium (b) Periosteum (c) Pleura (d) Pericardium

2. Basic structural unit of compact bone is: (a) Haversian system (b) Alveolus (c) Sarcomere (d) Nephron

3. Bone-resorbing cells are: (a) Osteocytes (b) Chondrocytes (c) Osteoblasts (d) Osteoclasts

4. Most common cartilage is: (a) Fibrocartilage (b) Calcified (c) Hyaline (d) Elastic

5. Most common freely movable joints are: (a) Sutures (b) Fibrous joints (c) Synovial joints (d) Cartilaginous joints

6. Functional unit of muscle contraction is: (a) Neuron (b) Nephron (c) Sarcomere (d) Osteon

7. Thick filaments of muscle are made of: (a) Myosin (b) Tropomyosin (c) Actin (d) Troponin

8. During contraction, calcium binds to: (a) Tropomyosin (b) Troponin (c) Myosin (d) Actin

Q.2 — SHORT QUESTIONS (6 × 2 = 12)

1. Differentiate between compact bone and spongy bone.

2. What are the functions of osteoblasts, osteocytes and osteoclasts?

3. Describe the types of cartilage and their locations.

4. What are the features of a synovial joint?

5. Differentiate between skeletal, smooth and cardiac muscles.

6. How do antagonistic muscles produce movement at a joint?

Q.3 — LONG QUESTIONS (2 × 8 = 16)

Q.3 (a) Describe the structure of bone and bone marrow. **(4)**

(b) Describe the types of cartilage and joints in the human body. **(4)**

Q.4 (a) Describe the ultrastructure of skeletal muscle. **(4)**

(b) Explain the sliding filament model of muscle contraction. **(4)**

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